



Department of Computer Science and Business Systems

Academic Year 2024– 2025 (Odd/~~Even~~ Semester)

Degree, Semester & Branch: B.Tech, V& CSBS

Course Code & Title: CCS335 & Cloud Computing

Name of the Faculty member(s): Dr. M. Gomathy Nayagam

Innovative Practice Description

- **Unit / Topic: Unit II/ Hypervisor**
- **Course Outcome : CO2**
- **Topic Learning Outcome: 2a**
- **Activity Chosen: Theory to Practice with Handson**
- **Justification:**

Using theory-to-practice active learning for the topic "Hypervisor" bridges theoretical concepts with hands-on application. Students first learn the underlying principles of hypervisors, such as their role in virtualization, types (Type 1 and Type 2), and their functions. Then, through practical exercises like setting up virtual machines, students directly apply this knowledge. This active learning approach reinforces theoretical concepts by allowing students to observe real-world implementations, troubleshoot issues, and engage in problem-solving. By connecting theory to practice, students gain a deeper, more practical understanding of hypervisor technology, enhancing retention and preparing them for real-world scenarios.

- **Time Allotted for the Activity : 45 Minutes**
- **Details of the Implementation:**
 - Course instructor explained the basic concept and he explained the usage of hypervisor and create VM on it with an example to the students for 30 Minutes in the class
 - After completion of the class, the course instructor shares the hypervisor software's to the students and he explained the step by step implementation of installing Hypervisor and create VM on it..
 - The fig. shows the sample glimpses of hands-on session in Cloud Computing laboratory

- The students installed the hypervisor software in their system and tries to create virtual machine over it with the specified configuration.

• **CO–PO/ PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 2	3	3	3	2	2	-	-	-	1	1	1	1	3	1	3
(1– Low			2–Moderate			3 – High)									

• **PO/ PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO5
	3	3	3	2	2
Justification for correlation	The students could be able to apply the basic engineering knowledge to install hypervisor and create VM over it.	The students could be able to formulate the solution for given problem using VM installed over hypervisor	The students could be able to provide the solution for identifying the relationship between the various hypervisors types	The students could be able to build the Virtual Machine in Hypevisor.	The students could be able to use the appropriate hypervisor softwares

Innovative practice	PO9	PO10	PO11	PO12	PSO1	PSO2
	1	1	1	1	3	1
Justification for correlation	The students could be able to know their responsibility to install hypervisor as an individual.	The students could be able to communicate their results of the problem and discuss with their peer team members	The students could be able to apply the various configuration features of hypervisors when they do mini project in this course	The students could be able to learn the newer hypervisor	The students could be able to apply VM technology	The students could be able to develop VM application

Innovative practice	PSO3
	3
Justification for correlation	The student could able to apply the concept of hypervisor in real-time project

• **Images/Screenshot of the practice:**

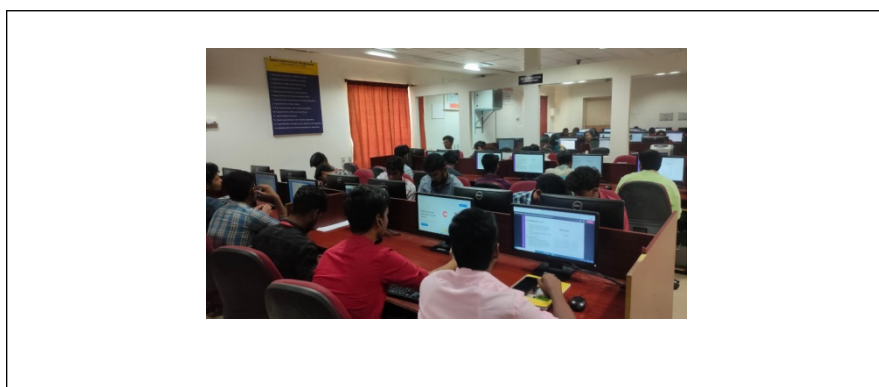


Fig.1. Glimpse of installing Hypervisor and Creating VM Hands-On session in Cloud Computing Laboratory

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

All the students are involved and worked in the hands-on session.

❖ **Benefit of the Practice (E.g. Outcome Attainment would have increased due to innovative practice over conventional practice)**

- It helps students to know the implementation of hypervisor.

❖ **Challenges faced in implementation:**

- Make all students to implement and correct the error during the installation.

References:

- ❖ Mr.M.Gomathy Nayagam, “OpenMP execution Model” in course CS6801 Multi-core Architecture and Programming, <https://www.ritrjpm.ac.in/images/computer-science/CS6801-MCAP-MGM.pdf>
- ❖ Dr.M. Gomathy Nayagam, “ Linear Least Square” in course AD3491 - Fundamentals of Data Science and Analytics. https://www.ritrjpm.ac.in/images/CSBS/2023-2024/TP_AD3491_FDSA_MGM_23_24.pdf

Signature of Faculty Member
(Dr.M.Gomathy Nayagam)

HOD